Building the Context Layer

Mode 2 Phase 1 (Building Blocks), Chapter 4/6 — Revision & Viva-Prep Cheatsheet (English + Roman Urdu)
Source: agentfactory.panaversity.org/docs/context-layer-crash-course

EN: A single Worker's store was a retrieval problem. A whole company's scattered sources (SharePoint, email, chat, live systems) is a GOVERNANCE problem wearing retrieval's clothes. Every returned item must answer 3 questions: Where did this come from? Can this person see it? Does it still apply?
Roman Urdu: Ek Worker ka store retrieval problem tha. Poori company ke scattered sources (SharePoint, email, chat, live systems) ek GOVERNANCE problem hai jo retrieval ke kapre pehne hai. Har return hone wala item 3 sawalon ka jawab de: Yeh kahan se aya? Kya yeh banda dekh sakta? Kya yeh abhi bhi lagu hota?

1. Four Source Classes

- (1) Agent Factory SoR -- shared method, cite-able. (2) Vertical SoRs -- customer's own rules, MCP-confirmed. (3) Customer operational records (ERP/CRM) -- live typed query, NEVER indexed. (4) Customer working context (email/chat) -- evidence only, NEVER a rule.
- Class 4 is the one people point to first -- and why 'fluent but baseless' answers are so common.

2. Permission Inheritance + the Gate

- Permission-sync/RBAC often isn't included in open-source community editions -- a real corpus must never be connected until the permission test passes.
- **Correct order:** resolve identity -> resolve permissions -> filter eligible docs (BEFORE retrieval) -> retrieve+rank -> assemble answer -> resolve action rights separately.
- Unsafe order: retrieve everything, then tell the model 'don't mention what you shouldn't see' -- a hidden passage inside the model's context is NOT actually hidden.

3. Two-Call Pattern + Authority Routing

- **Discovery** (where a rule might be, a pointer) is separate from **Confirmation** (officially applicable right now -- domain/class/jurisdiction/version/approval checked).
- Live state (balances, approvals, current versions) is NEVER indexed -- indexing it makes a stale copy. Working context gets indexed, governed knowledge gets discovered, current truth gets live-queried.

4. Provenance + Conflict + the 8 Rules

- Every item carries a citation envelope: source system, stable ID, authority class, version+period, permission basis. Compression kills provenance -- compress prose, never provenance.
- **Conflict has 3 valid outcomes:** resolved by scope, resolved by authority, unresolved-escalate. A 4th (blending sources into one smooth sentence) is never acceptable.

Key Terms Cheatsheet

One line each — English + Roman Urdu.

Term	English	Roman Urdu
4 Source Classes	Agent Factory SoR, Vertical SoRs, Operational records, Working context.	Agent Factory SoR, Vertical SoRs, Operational records, Working context.
Two-Call Pattern	Discovery (where it might be) is separate from Confirmation (officially applicable now).	Discovery (kahan ho sakta) Confirmation (abhi officially applicable) se separate hai.
Permission-before-retrieval	Filtering eligible docs BEFORE retrieval, not hiding them after.	Eligible docs RETRIEVAL SE PEHLE filter karna, baad mein hide karna nahi.
Citation envelope	Source, stable ID, authority class, version+period, permission basis -- travels with every item.	Source, stable ID, authority class, version+period, permission basis -- har item ke sath travel karta.
3 valid conflict outcomes	Resolved by scope, resolved by authority, unresolved-escalate -- never blended.	Resolved by scope, resolved by authority, unresolved-escalate -- kabhi blend nahi.

Quick Self-Check

Question	Answer
Why is Class 4 (working context) the riskiest source class?	It carries evidence, never professional authority -- yet it's where people most often point first, producing fluent-but-baseless answers.
Why must permission filtering happen BEFORE retrieval, not after?	A passage the model already sees in its context isn't actually hidden, no matter what instruction tells it not to mention it.
What's the difference between Discovery and Confirmation?	Discovery finds where a rule MIGHT be (a pointer); Confirmation checks whether it's officially applicable right now (domain/jurisdiction/version/approval).
Why is live operational state (balances, current versions) never indexed?	Indexing it creates a stale copy -- it must always be a live typed query to the owning system instead.
What happens when two sources conflict?	It must be resolved by scope, resolved by authority, or escalated unresolved -- never silently blended into one smooth answer.

Where This Leads / Sources

Leads to: Eval-Driven Development, Building a Digital FTE (knowledge-access bridge). Full Northstar worked case + 5 failure tests in Part 4.

Source: docs/context-layer-crash-course/ (this repo), content originally via Zia Tutor AI connector.